

BEDDING PLANE The surface separating a layer of a sedimentary rock from the layer above or below. See also *bedding*, *sedimentary rock*.

BERGSCHRUND A deep crack or crevasse that forms where the moving ice of a glacier pulls away from its headwall. See also *crevasse* (panel), *glacier*.

BERM A ridge of shingle on the upper part of a beach, above the beach face, usually marking the point of the highest high tides. See also *beach face*.

BIODIVERSITY The variety of living plants and animals in a given area or habitat.

BIOLUMINESCENCE The production of light by living life forms. Bioluminescent organisms include some bacteria, fungi, cephalopods, jellyfish, and fish.

BIOME A large-scale biological community defined primarily by its vegetation. Examples include tropical rain forest, temperate grassland, taiga, and tundra.

BLACK SMOKER A hydrothermal vent in which the emerging, very hot water is colored black with dark

minerals—mostly iron sulfide. See also *hydrothermal vent*, *white smoker*.

BLOCK LAVA Volcanic lava with a surface of smooth-sided angular fragments. This lava usually solidifies to form the rock andesite.

BLOCKFIELD An area of broken rocks, usually in a mountainous region. Also called *felsenmeer*.

BOG A peat-accumulating wetland. Bogs are acidic, with low levels of nutrients, and their soils are composed almost entirely of dead plant matter. See also *fen*, *marsh*.

BORE See *tidal bore*.

BOREAL A term used for parts of the Northern Hemisphere between the Arctic and temperate zones—for example, the boreal forest of Eurasia.

BOSS An igneous intrusion that is roughly circular in horizontal cross section. See also *igneous intrusion* (panel).

BOUNDARY See panel, below.

BRACKISH Water that has a salt content less than that of ocean water but greater than that of freshwater.

BRAIDED STREAM A watercourse where water flows in several shallow channels that constantly separate and rejoin. This system is common downstream of many glaciers. See also *glacier*, *meltwater*.

BREAKER ZONE The part of a beach where waves break. See also *beach* (panel), *beach face*, *swash*.

BRECCIA A sedimentary rock made up of angular fragments of minerals and other rocks.

BROADLEAF A term describing trees that have broad leaves, as distinct from needlelike leaves. Examples of broadleaved trees include chestnuts, maples, and oaks. See also *coniferous*, *deciduous*, *evergreen*.

C

CAATINGA A kind of thorny forest found in semiarid areas of northeast Brazil.

CALCITE A common transparent or opaque mineral form of calcium carbonate.

CALDERA A large, bowl-shaped volcanic depression, typically more than 0.6 mile (1 km) in diameter. It forms when a volcano collapses into its magma chamber after it has been emptied by a volcanic explosion. See also *magma chamber*, *volcano* (panel).

CALVING The process of ice blocks shedding from a glacier or ice sheet and falling into a lake or the ocean. See also *glacier*, *ice sheet*.

CANOPY The crowns of the tallest forest trees, which receive full sunlight. The canopy may be closed, with crowns touching each other to form a continuous layer, or open. See also *rain forest*.

CANYON A deep, steep-sided, relatively narrow valley. See also *gorge*.

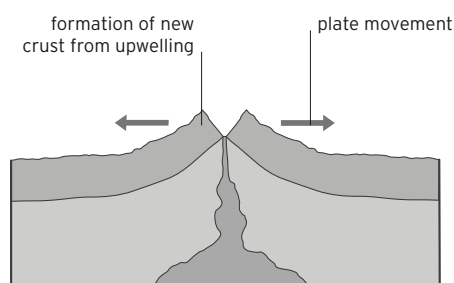
CATCHMENT AREA The total area drained by one river and its tributaries. See also *drainage basin*, *watershed*.

CERRADO A type of savanna dotted with small trees and shrubs, found mostly in central Brazil. See also *savanna*.

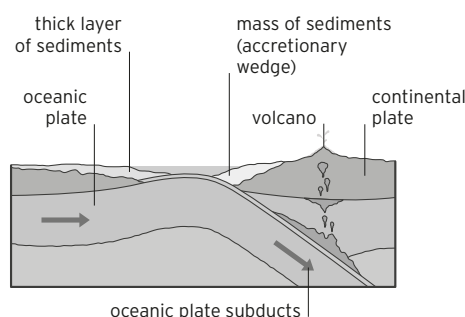
BOUNDARY

The margin between two or more tectonic plates is called a boundary. At a divergent (constructive) boundary, two plates move apart, creating a rift. Magma rises from the mantle beneath the rift, pouring out through long fissures. A convergent (destructive) boundary forms where two plates collide, often where oceanic crust is pushing against continental crust and is being forced

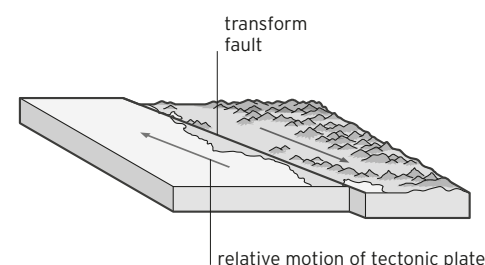
beneath it, a process known as subduction. The continental crust often buckles up into a mountain range, while the subducting oceanic crust forms a deep seafloor trench. Volcanoes form above the area of subduction. A transform plate boundary involves two plates pushing past each other. Although no magma reaches the surface, earthquakes are regular features of transform fault locations.



DIVERGENT (CONSTRUCTIVE) BOUNDARY



CONVERGENT (DESTRUCTIVE) BOUNDARY



TRANSFORM (CONSERVATIVE) BOUNDARY